Day3 演習1

```
import matplotlib.pyplot as plt
import numpy as np
from scipy.integrate import solve_ivp

def selkov_model( t , init , *param ):
    x = init[ 0 ]
    y = init[ 1 ]

    [ a , b ] = param


    v1 = b
    v2 = a * y + (x ** 2) * y
    v3 = x

    dxdt = v2 - v3
    dydt = v1 - v2

    return [ dxdt , dydt ]


init = [ 1.0 , 1.0 ]
t_span = [ 0 , 100 ]
t_eval = np.linspace( t_span[0] , t_span[1] , 10000 )

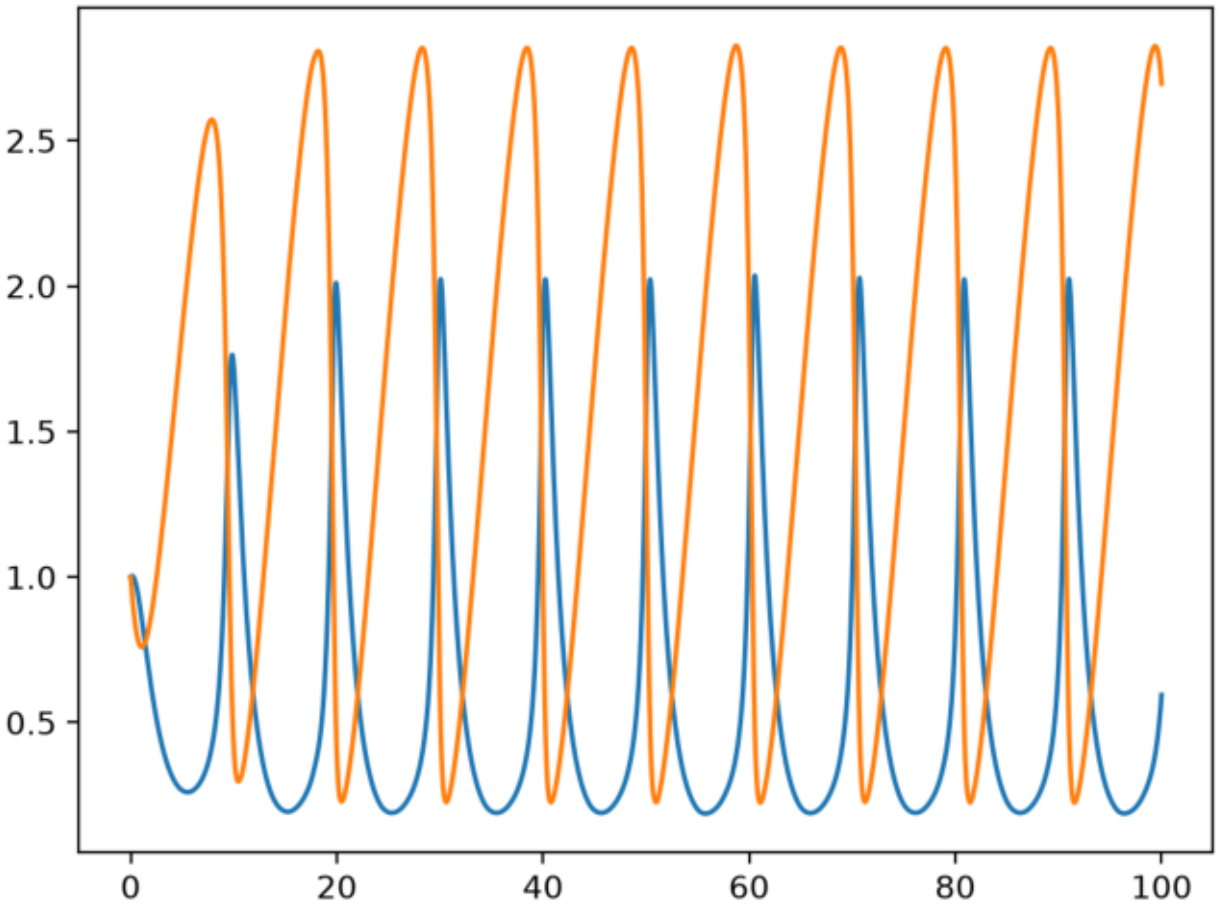
a = 0.06
b = 0.6

param = [ a , b ]

solution = solve_ivp( selkov_model, t_span, init, t_eval=t_eval, method='RK45',
                      args=param )

ADP = solution.y[0]
F6P = solution.y[1]

plt.plot( solution.t , ADP )
plt.plot( solution.t , F6P )
plt.show()
```


2.5
2.0
1.5
1.0
0.5
0
20
40
60
80
100